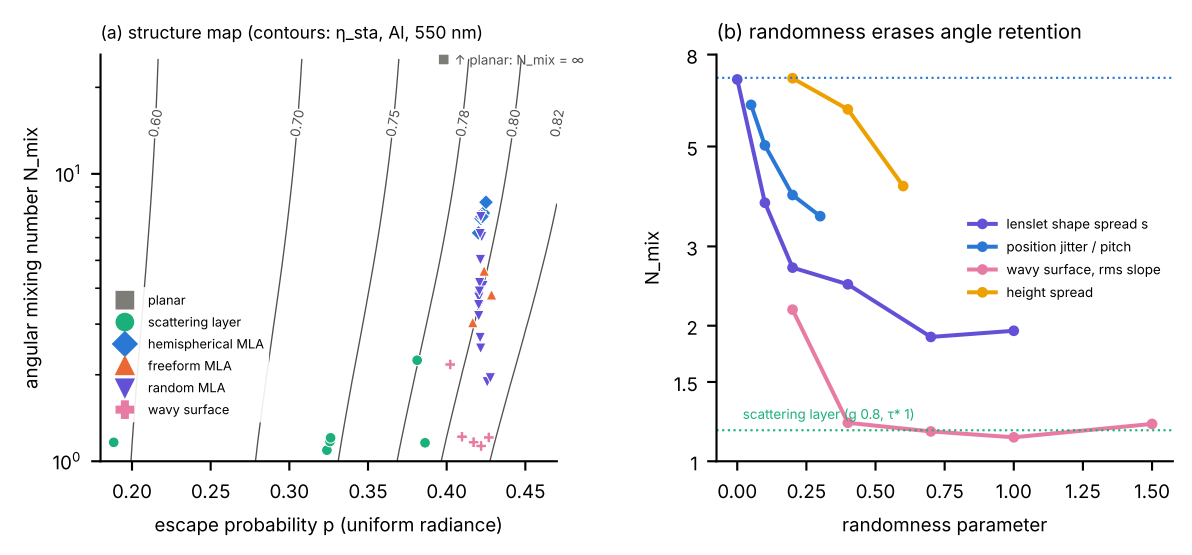
(a) structure map (contours: η_sta, AI, 550 nm)
↑ planar: N_mix = ∞
0.60
0.70
0.75
0.78
0.80
0.82
planar
scattering layer
hemispherical MLA
freeform MLA
random MLA
wavy surface
angular mixing number N_mix
10^1
10^0
0.20
0.25
0.30
0.35
0.40
0.45
escape probability p (uniform radiance)
(b) randomness erases angle retention
N_mix
8
5
3
2
1.5
1
lenslet shape spread s
position jitter / pitch
wavy surface, rms slope
height spread
scattering layer (g 0.8, τ* 1)
0.00
0.25
0.50
0.75
1.00
1.25
1.50
randomness parameter